

23 Co-Optimization of Battery Storage Between Energy Arbitrage and Frequency Reserve Markets

Context

Batteries can provide both energy arbitrage (buy low, sell high) and frequency reserves. Co-optimizing participation in these markets can significantly affect revenues and risk.

Objectives

- Model a battery capable of both energy arbitrage and reserve provision.
- Formulate a co-optimization problem deciding how much capacity to allocate to each service.
- Evaluate revenue and risk under historical price and reserve remuneration data.

Data

- Historical hourly day-ahead prices,
- Historical reserve prices or capacity payments for frequency regulation / primary or secondary reserve (see <https://www.services-rte.com/en/view-data-published-by-rte/balancing-capacities.html>),
- Battery technical parameters (capacity, power, efficiency).

Suggested Tasks

1. Specify the battery model with constraints on energy and power.
2. Define decision variables:
 - Hourly charging/discharging for arbitrage,
 - Reserve capacity allocated in each hour (or block).
3. Formulate an optimisation problem over a representative horizon:
 - Maximise total revenue from arbitrage + reserve capacity payments,
 - Subject to battery constraints and possibly minimum state-of-charge for reserve provision.
4. Implement and solve the problem using `cvxpy` or similar.
5. Compare:
 - Pure arbitrage vs pure reserve vs co-optimised solution,
 - Revenues, utilisation and risk characteristics.